Code No: R07A1EC01 **Set No. 1**

I B.Tech Semester Regular Examinations, June 2009
C PROGRAMMING AND DATA STRUCTURES
(Common to Civil Engineering, Electrical & Electronic Engineering, Electronics & Communication Engineering, Computer Science & Engineering, Chemical Engineering, Electronics & Instrumentation Engineering, Bio-Medical Engineering, Information Technology, Electronics & Control Engineering, Computer Science & Systems Engineering, Electronics & Telematics, Electronics & Computer Engineering, Aeronautical Engineering, Instrumentation & Control Engineering and Bio-Technology)

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks
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1. (a) Define Algorithm.
   (b) What is the use of flowchart ?
   (c) What are the different steps followed in the program development? [3+3+10]
2. (a) What is the advantage of using header files in 'C' ?
   (b) Write a program to generate Fibonacci series using recursion. [6+10]
3. What are the various operations performed in pointers explain with an example? [16]
4. Define Structure and write the general format for declaring and accessing members. [16]
5. What are the different Input/Output operations on Files ? [16]
6. Write a program to explain insertion sort . Which type of technique does it belong. [16]
7. List the advantages of doubly linked list . Write a program in C to perform the following operations in a doubly linked list.
   (a) Search
   (b) print list forward
   (c) Print list reverse [16]
8. (a) What are the differences between a tree and binary tree?
   (b) Give the representation of binary tree and explain [8+8]

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Code No: R07A1EC01

Set No. 2

I B.Tech Semester Regular Examinations, June 2009
C PROGRAMMING AND DATA STRUCTURES
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Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Define Algorithm.
   (b) What is the use of flowchart ?
   (c) What are the different steps followed in the program development? [3+3+10]
2. What is a function ? What are the different types of functions? Explain function with no argument and no return type with an example. [16]
3. How multidimensional array is passed to function , how are the formal argument declaration written ? Compare with one dimensional array . [16]
4. What is a self - referential structure? What kind of applications are it is used? give example. [16]
5. (a) List and explain the operations of file ?
   (b) What are the different functions of file ? [8+8]
6. Write a program to explain insertion sort . Which type of technique does it belong. [16]
7. Write a program to evaluate the following expression A / B * (C + D) / A to prefix using stack. [16]
8. (a) What is a network?
   (b) What is a spanning tree?
   (c) Define minimal spanning tree .
   (d) What are the various traversals in a tree? [16]

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Code No: R07A1EC01 **Set No. 3**

I B.Tech Semester Regular Examinations, June 2009
C PROGRAMMING AND DATA STRUCTURES
(Common to Civil Engineering, Electrical & Electronic Engineering, Electronics & Communication Engineering, Computer Science & Engineering, Chemical Engineering, Electronics & Instrumentation Engineering, Bio-Medical Engineering, Information Technology, Electronics & Control Engineering, Computer Science & Systems Engineering, Electronics & Telematics, Electronics & Computer Engineering, Aeronautical Engineering, Instrumentation & Control Engineering and Bio-Technology)

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks
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1. (a) Define Algorithm.
   (b) What is the use of flowchart ?
   (c) What are the different steps followed in the program development? [3+3+10]
2. What are the facilities provided by the preprocessor statement. Explain with a program. [16]
3. How multidimensional array is passed to function , how are the formal argument declaration written ? Compare with one dimensional array . [16]
4. Define Structure and write the general format for declaring and accessing members. [16]
5. How the data is searched in sequential files. Mention the different techniques used with an example ? [16]
6. Write a program to explain bubble sort. Which type of technique does it belong. [16]
7. Explain the tower-of-hanoi problem. [16]
8. (a) Define tree. What is a subtree?
   (b) Define the following terms.children nodes, siblings, root node, leaves level and degree of tree . [8+8]

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Code No: R07A1EC01 **Set No. 4**

I B.Tech Semester Regular Examinations, June 2009
C PROGRAMMING AND DATA STRUCTURES
(Common to Civil Engineering, Electrical & Electronic Engineering, Electronics & Communication Engineering, Computer Science & Engineering, Chemical Engineering, Electronics & Instrumentation Engineering, Bio-Medical Engineering, Information Technology, Electronics & Control Engineering, Computer Science & Systems Engineering, Electronics & Telematics, Electronics & Computer Engineering, Aeronautical Engineering, Instrumentation & Control Engineering and Bio-Technology)

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks
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1. (a) Write a 'C' program which reads temperature in either Fahrenheit or Celsius and compute temperature in the opposite scale.
   Formulas are C = (F - 32) * 5/9 , F = 9 (C/5) + 32
   (b) Write a 'C' program to find greatest common division (GCD) of two given numbers. [6+10]
2. Explain the following library functions
   (a) sqrt(x)
   (b) fmod(x,y)
   (c) toupper(x)
   (d) fmod(x,y) [16]
3. Define an array. What are the different types of arrays. Explain. [16]
4. Define Structure and write the general format for declaring and accessing members. [16]
5. (a) Write the syntax for opening a file with various modes and closing a file .
   (b) Explain about file handling functions . [8+8]
6. Write a program to explain selection sort . Which type of technique does it belong. [16]
7. Define doubly linked list . Write a program in C to perform the following operations in a doubly linked list.
   (a) Insert
   (b) Delete
   (c) Display [16]

8. Write a 'C' program to implement recursive algorithm for a Binary Search Tree. [16]

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